



BUSINESS DATA MANAGEMENT CAPSTONE PROJECT

**Project on:
Data-Driven Sales, Inventory, and
Profit Optimization in
Agricultural Retail Business**

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Organization Background

Krishi Khad Bhandar

Agricultural retail shop located in Belhariya, Gaya, Bihar

Established 20 years ago

Business Type : B2B and B2C both

Supplies:

- Fertilizers

- Seeds

- Pesticides

Serves:

- Farmers

- Local retailers

Dataset used:

- Sales records from March 2023 - December 2025

Key business challenges:

- Difficulty understanding seasonal sales trends

- Inventory management issues

- Uneven regional revenue distribution

- Storage space constraints



Data Collection

Dataset used in the project includes sales records from 2023-2025.

Variables collected:

- Transaction Date
- Product Name
- Product Category
- Unit Price
- Quantity Sold
- Total Sales Value
- Region

Data Source:

Shop transaction records and sales registers.

date	Region	Product	Category	Weight pe	Price per l	Quantity	Total Valu
9/11/2023	Jamuhar	cropyte	pesticide	50g	115	20	2300
9/11/2023	Jamuhar	action	pesticide	25 g	62	10	620
9/11/2023	Jamuhar	Action	pesticide	100g	230	5	1150
9/8/2023	Jamuhar	Action	Pesticide	25g	62	20	1240
9/8/2023	Jamuhar	Action	Pesticide	25g	62	20	1240
9/8/2023	Jamuhar	Mahajivan	Fertilizer	1 kg	90	10	900
9/8/2023	Jamuhar	Mahapow	Fertilizer	1kg	130	10	1300
9/8/2023	Jamuhar	Sticker	Pesticide	100g	80	12	960
9/8/2023	Jamuhar	Cropyte	Pesticide	50g	115	24	2760
9/8/2023	Jamuhar	Mahakal	Pesticide	20g	80	5	400
9/8/2023	Barachatti	Palak	Seed	1 kg	110	10	1100
9/8/2023	Barachatti	coriander	Seed	1kg	220	10	2200
9/8/2023	Barachatti	Coriander	Seed	1 kg	300	2	600
10/18/2023	Barachatti	Mustard	Seed	1kg	230	30	6900
10/18/2023	Barachatti	Cobra	Pesticide	1kg	40	25	1000

Data Cleaning & Preparation

Before analysis the dataset was processed using Microsoft Excel.

Data preprocessing steps:

- Removing duplicate entries
- Handling missing values
- Standardizing product names
- Organizing dataset into structured format

Purpose:

Ensure accurate and reliable analysis.

Tools used:

- Excel filtering
- Sorting
- Pivot tables
- Conditional formatting

Problem 1: Difficulty in Understanding Seasonal Sales Trends

Background:

Although the store has accumulated several years of sales data, it is not utilized effectively for decision-making. Specifically, the store cannot identify which season generates the highest revenue.

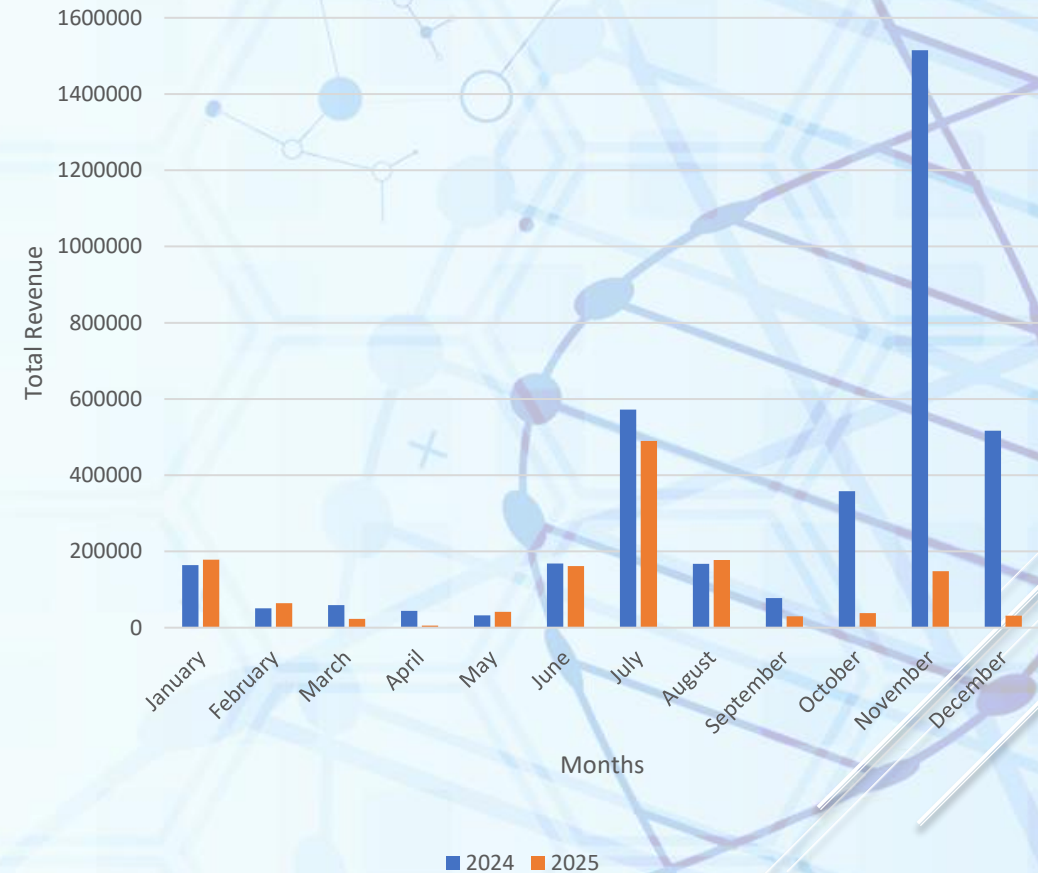
Objective:

- Identify sales trends and seasonal demand patterns using trend analysis.

Key Insights:

- Sales show a strong seasonal pattern.
- Peak demand in July (Kharif season) and Nov-Dec (Rabi season).
- Demand for seeds and fertilizers follows crop sowing cycles.

Monthly Revenue Sales (2024 vs 2025)



Problem 2 : Inventory management issues

Background:

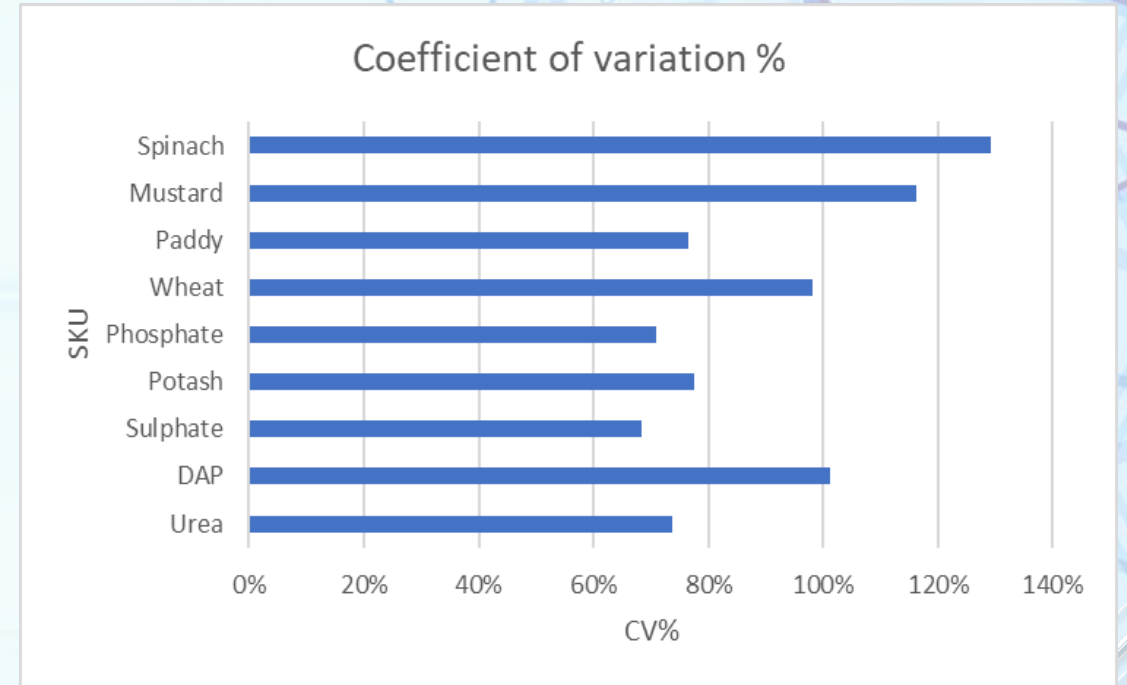
Agricultural demand is highly seasonal, causing large fluctuations in product sales and making inventory planning difficult.

Objective:

Evaluate demand variability across products using the Coefficient of Variation (CV) and determine appropriate Safety Stock levels.

Key Insights:

- Several products show high demand variability, increasing inventory risk.
- Spinach, Mustard, and DAP have very high CV values, indicating unpredictable demand.
- High-demand products like Paddy and Urea require larger safety stock to prevent stockouts during peak seasons.
- Stable products such as Sulphate and Phosphate require lower buffer inventory.



Problem 3: Uneven Regional Revenue Distribution

Background:

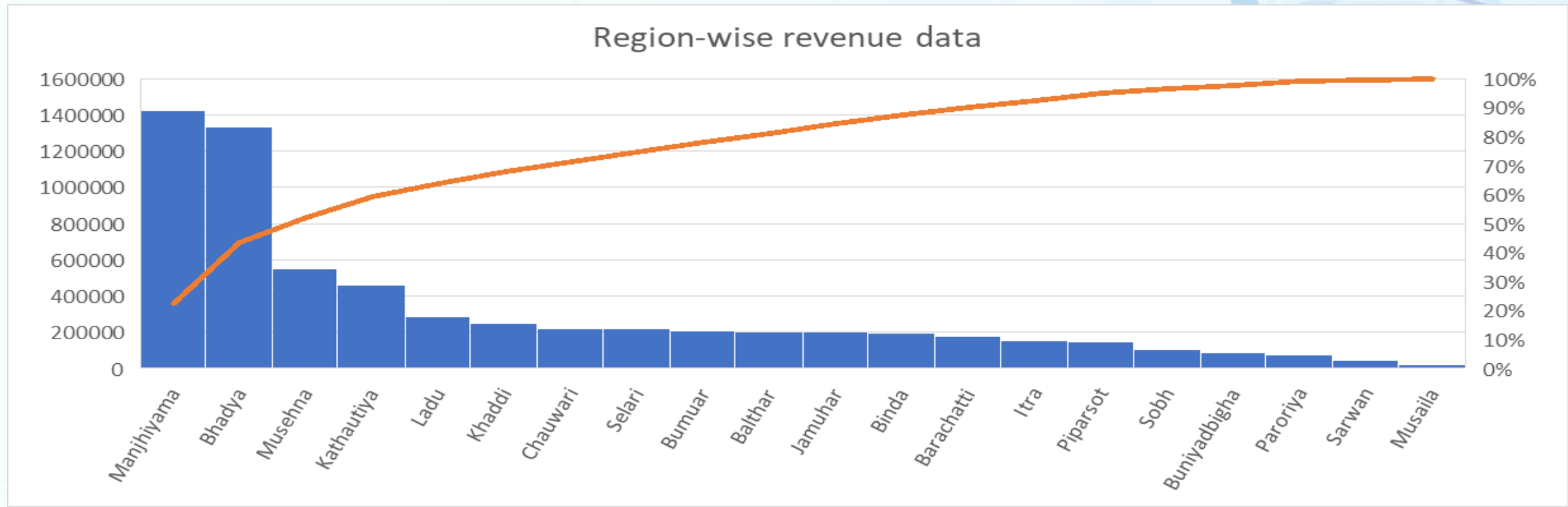
Lack of regional sales analysis makes it difficult to plan inventory, manage storage space, and understand differences in demand across villages.

Objective:

Analyze regional revenue contribution to identify high-performing and low-performing markets.

Key Insights:

- Revenue is highly concentrated in a few regions.
- **Manjhiyama** and **Bhadya** contribute the highest share of total revenue.
- Several regions such as **Sarwan**, **Musaila**, **Paroriya**, and **Buniyadbigha** generate relatively low sales.
- Overdependence on a few regions increases business risk if demand declines in those areas.



Problem 4: Limited Storage Space

Background:

Limited storage space in the shop leads to inefficient stock management, especially during peak agricultural seasons when multiple fertilizers arrive simultaneously.

Objective:

Understand relationships between products using correlation analysis to optimize storage and stocking strategies.

Key Insight : Complementary Product Demand

- Strong positive correlations exist between Urea, Potash, Sulphate, and DAP.
- Farmers often purchase these fertilizers together during the same crop cycle.
- This indicates complementary demand among these products.

	DAP	Pest control	Phosphate	Potash	Sulphate	Urea	Zyme
DAP	1	0.0338414	0.36429796	0.67499	0.8894131	0.70845	0.40309535
Pest control	0.0338414	1	-0.4643789	-0.4966	-0.159499	-0.21764	0.01405145
Phosphate	0.36429796	-0.464379	1	0.72844	0.4609737	0.70258	0.05477807
Potash	0.67498543	-0.496604	0.72844459	1	0.8839244	0.96148	0.17500855
Sulphate	0.88941311	-0.159499	0.46097374	0.88392	1	0.88045	0.47840726
Urea	0.70844614	-0.217643	0.70258469	0.96148	0.8804517	1	0.22953153
Zyme	0.40309535	0.0140514	0.05477807	0.17501	0.4784073	0.22953	1

Business Recommendations

Difficulty in Understanding Seasonal Sales Trends

- Seasonal Inventory Planning
- Increase stock before June–July (Kharif season).
- Increase stock before October–November (Rabi season).
- Reduce inventory during March–May off-season.

Inventory management issues

- Implement seasonal inventory planning by increasing stock before major sowing seasons.
- Maintain higher safety stock for high-CV products to avoid stockouts.
- Use moderate inventory levels for medium variability products and regular replenishment for stable products.
- Monitor demand patterns regularly to adjust stock levels and reduce excess inventory.

Uneven Regional Revenue Distribution

- Maintain higher inventory availability in high-performing regions to avoid stock shortages.
- Implement targeted promotions and awareness campaigns in medium-performing regions.
- Explore distribution expansion and improved accessibility in low-performing regions.
- Diversify sales across more villages to reduce revenue concentration risk.

Limited Storage Space

- Bundle-Based Product Strategy:
- Stock Urea, Potash, Sulphate, and DAP together to improve storage efficiency.
- Offer fertilizer bundle deals during peak farming seasons.
- Arrange complementary products in the same storage or display area to simplify inventory management and improve sales.

Improvement & Future Growth



Inventory Automation

Adopt basic inventory management software to track stock levels and reduce shortages.



Demand Forecasting

Use time-series forecasting models to predict seasonal demand more accurately.



Regional Market Expansion

Expand sales in low-performing regions through awareness campaigns and better distribution.



Digital Sales & Delivery

Introduce phone/WhatsApp ordering and home delivery for farmers or local retailers in nearby villages.



Data-Driven Decision Making

Continuously analyze sales data to improve inventory planning, marketing, and storage management.

CONCLUSION

Project Impact

The recommendations provided are simple, practical, and data-driven steps, not complex theories
By implementing these strategies agricultural store can move from state of reaction to one of control.

Thank You